



SCEGGS Darlinghurst

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Centre Number

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Student Number

2009

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1–7
- All questions are of equal value

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Total marks – 84

Attempt Questions 1–7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

(a) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ 1

(b) Find the co-ordinates of the point P which divides AB externally in the ratio $2:3$ where $A(1, -4)$ and $B(6, 9)$. 2

(c) Solve for x : 3

$$\frac{4}{x-1} \geq 1$$

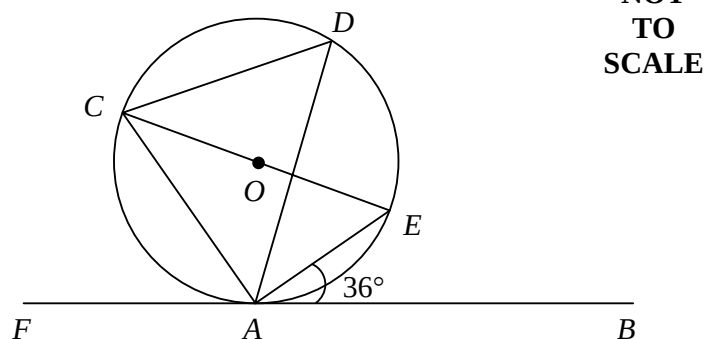
(d) The angle between two lines $y = mx$ and $y = \frac{1}{3}x$ is $\frac{\pi}{4}$. 2
Find the exact values of m .

(e) If α , β and γ are the roots of $x^3 - 3x + 5 = 0$ find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. 2

(f) Use the table of standard integrals to find $\int \sec 2x \tan 2x \, dx$. 2

Question 2 (12 marks) Use a SEPARATE writing booklet.

(a)



FB is a tangent meeting a circle at A . CE is the diameter, O is the centre and D lies on the circumference. $\angle BAE = 36^\circ$.

(i) Find the size of $\angle ACE$, giving reasons. 1

(ii) Find the size of $\angle ADC$, giving reasons. 2

(b) Find $\int \frac{dx}{\sqrt{25 - 4x^2}}$ 2

(c) (i) If $\sin x - \sqrt{3} \cos x = R \sin(x - \alpha)$ find R and α where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2

(ii) Find the general solution for $\sin x - \sqrt{3} \cos x = \sqrt{2}$ (leave your answer in exact form). 2

(d) Colour blindness affects 5% of all men. What is the probability that any random sample of 20 men should contain:

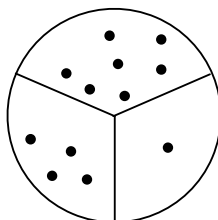
(i) no colour blindness. 1

(ii) two or more colour blind men (to 3 decimal places). 2

Question 3 (12 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_0^{\frac{3}{2}} \sqrt{9 - x^2} \, dx$ using the substitution $x = 3 \sin \theta$. 3

- (b) Twelve points lie inside a circle. No three points are collinear. Seven of the points lie in sector 1, four lie in sector 2 and the other point lies in sector 3.



- (i) Show that 220 triangles can be made using these points. 1
- (ii) One triangle is chosen at random from all possible triangles. Find the probability that the triangle chosen has one vertex in each sector. 1
- (iii) Find the probability that the vertices of the triangle chosen all lie in the same sector. 1
- (c) (i) Sketch the graph of the function $f(x) = e^x - 2$. 1
- (ii) On the same diagram sketch the graph of the inverse function $f^{-1}(x)$. 1
- (iii) State the equation of the function $f^{-1}(x)$. 1
- (iv) Explain why the x co-ordinate of any point of intersection of the graphs $y = f(x)$ and $y = f^{-1}(x)$ satisfies the equation $e^x - x - 2 = 0$. 1
- (v) One root of the equation $e^x - x - 2 = 0$ lies between $x = 1$ and $x = 2$. Use one application of Newton's method, with a starting value of $x = 1.5$, to approximate the root, to 2 decimal places. 2

Question 4 (12 marks) Use a SEPARATE writing booklet.

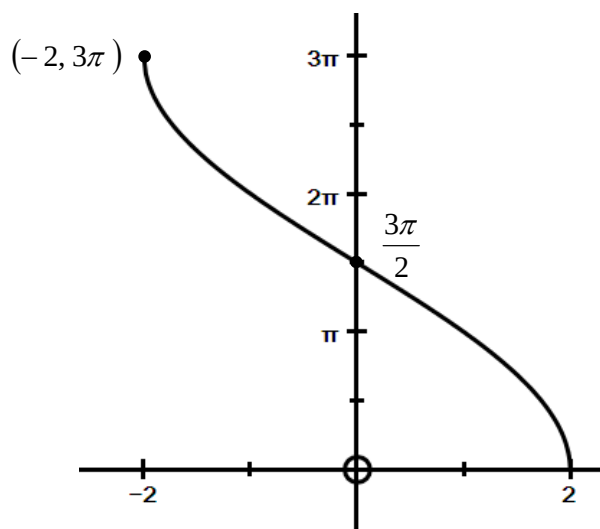
- (a) $P(4p, 2p^2)$ is a variable point on the parabola $x^2 = 8y$.
The normal at P cuts the y -axis at A and R is the midpoint of AP .

(i) Show that the normal at P has equation $x + py = 4p + 2p^3$. 2

(ii) Show that R has co-ordinates $(2p, 2p^2 + 2)$. 2

(iii) Show that the locus of R is a parabola and find its vertex and focus. 3

- (b) The graph of $y = a \cos^{-1} bx$ is drawn below.

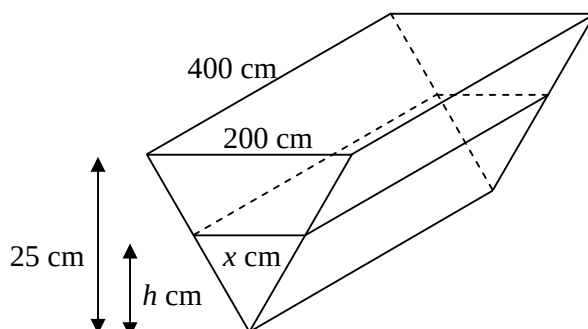


(i) Find a and b . 2

(ii) Find the exact area bound by the curve and the y -axis for $0 \leq y \leq \frac{\pi}{2}$. 3

Question 5 (12 marks) Use a SEPARATE writing booklet.

(a)



An open, flat topped water trough is in the shape of a triangular prism.

Its rectangular top measures 200 cm by 400 cm and its triangular cross-section has a vertical height of 25 cm.

When the water depth is h cm the water surface measures x cm by 400 cm.

- (i) Show that when the water depth is h cm the volume $V \text{ cm}^3$ of water in the trough is given by $V = 1600h^2$. 2

Water is being emptied through a hole in its base at a constant rate of 16 L per second.

- (ii) Find the rate at which the depth of water is changing when $h = 10$ cm. 2

Question 5 continues on page 7

Question 5 (continued)

- (b) After cooking her cheesecake, Donna puts it in the fridge. The fridge is running at a constant temperature of 8°C . At time t minutes the temperature T of the cheesecake decreases according to the equation:

$$\frac{dT}{dt} = -k(T - 8) \text{ where } k \text{ is a positive constant.}$$

Donna puts the cheesecake in the fridge at 9.00am when its temperature is 85°C .

- (i) Show that $T = 8 + 77e^{-kt}$ satisfies both this equation and the initial conditions. 2

- (ii) Donna checks the temperature of the cheesecake at 10.00am and it is 40°C . 3

It is best served when it reaches a temperature of 10°C .

At what time (to the nearest minute) should Donna serve the cheesecake?

- (c) In the expansion of $(1 + ax)^{10}$, the coefficient of x^6 is twice the coefficient of x^7 . 3

Find the value of a .

End of Question 5

Question 6 (12 marks) Use a SEPARATE writing booklet.

- (a) (i) In the expansion of $(a + b)^{20}$ show that $\frac{T_{n+1}}{T_n}$ is given by: 2

$$\frac{21-n}{n} \frac{b}{a}$$

(where $T_{n+1} = {}^{20}C_n a^{20-n} b^n$)

- (ii) In the game of craps, 2 dice are thrown and the score is recorded as the sum of the uppermost faces of the dice.

α) Find the probability that a score of 7 is recorded. 1

β) If two dice are rolled 20 times, what is the most probable number of scores of 7 thrown? Calculate the probability that this occurs. 3

- (b) (i) Use the method of mathematical induction to show that if x is a positive integer then $(1 + x)^n - 1$ is divisible by x for all positive integers $n \geq 1$. 3

(ii) Factorise $12^n - 4^n - 3^n + 1$. 1

(iii) Without using the method of mathematical induction, deduce that $12^n - 4^n - 3^n + 1$ is divisible by 6 for all positive integers $n \geq 1$. 2

Question 7 (12 marks) Use a SEPARATE writing booklet.

(a) (i) Show that in the expansion of $\left(x - \frac{1}{x}\right)^{2n}$, the term independent of x is $(-1)^n \times {}^{2n}C_n$. **2**

(ii) Show that $\left(1 + x\right)^{2n} \left(1 - \frac{1}{x}\right)^{2n} = \left(x - \frac{1}{x}\right)^{2n}$ **1**

(iii) Deduce that:

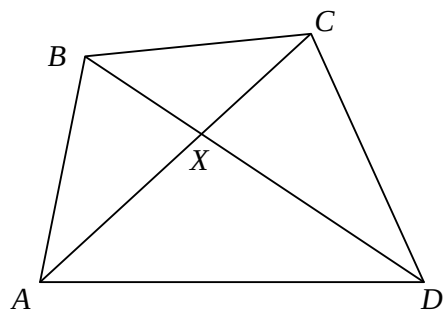
$$\left({}^{2n}C_0\right)^2 - \left({}^{2n}C_1\right)^2 + \left({}^{2n}C_2\right)^2 - \dots + \left({}^{2n}C_{2n}\right)^2 = (-1)^n \times {}^{2n}C_n$$
 2

Question 7 continues on page 10

Question 7 (continued)

(b) (i)

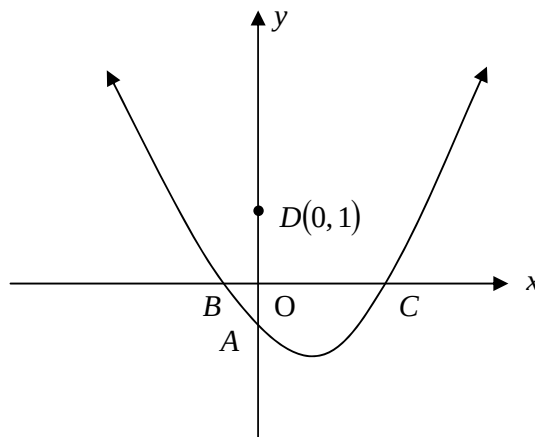
4



In the diagram above, the diagonals of quadrilateral $ABCD$ intersect at X .

Show that if $AX \cdot XC = BX \cdot XD$, $ABCD$ is a cyclic quadrilateral.

(ii)



Consider the parabola $y = x^2 + px - q$, where $q > 0$.

Let the parabola intercept the y-axis at A and the x-axis at the distinct points B and C .

D is the point $(0, 1)$

α) Find the co-ordinates of B and C .

1

β) Show that $ABDC$ is a cyclic quadrilateral.

2

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

Extension 1 Mathematics Trial HSC 2009 - Solutions

Q1 a) $\lim_{n \rightarrow 0} \frac{\sin 3n}{5n} = \frac{3}{5} \lim_{n \rightarrow 0} \frac{\sin 3n}{3n}$

$$= \frac{3}{5} \times 1 \quad \checkmark$$

$$= \frac{3}{5}$$

b) A(1, -4) B(6, 9)

$$\begin{array}{c} \nearrow \\ -2:3 \end{array}$$

$$x = \frac{3 \times 1 + -2 \times 6}{-2 + 3} \quad y = \frac{3 \times -4 + -2 \times 9}{-2 + 3} \quad \checkmark$$

$$= \frac{-9}{1}$$

$$= -9$$

$$= \frac{-30}{1}$$

$$= -30$$

$$P(-9, -30) \quad \checkmark$$

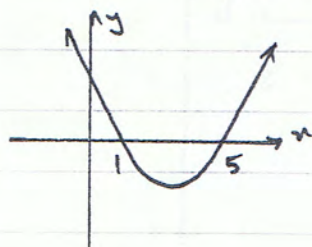
c) $(n-1)^2 \frac{4}{n-1} \geq 1 (n-1)^2$

$$4(n-1) \geq n^2 - 2n + 1$$

$$4n - 4 \geq n^2 - 2n + 1$$

$$0 \geq n^2 - 6n + 5 \quad \checkmark$$

Sketch $y = n^2 - 6n + 5$
 $= (n-5)(n-1)$



$$1 < n \leq 5 \quad \checkmark$$

Reas 3

d) $\left| \frac{n - \frac{1}{3}}{1 + n \times \frac{1}{3}} \right| = \tan \frac{\pi}{4}$

$$\left| \frac{n - \frac{1}{3}}{1 + \frac{n}{3}} \right| = 1 \quad \checkmark$$

$$\left| n - \frac{1}{3} \right| = \left| 1 + \frac{n}{3} \right|$$

Don't call

Those who used the 'cross' method were more successful than those who used formulae.

- Silly algebraic ~~mistake~~ were made here
 Take your time with Q1
 Also $x \neq 1$

You need to remember exact values of trig ratios. i.e. $\tan \frac{\pi}{4} = 1$ not $\frac{1}{\sqrt{2}}$

$$M - \frac{1}{3} = 1 + \frac{M}{3}$$

$$3M - 1 = 3 + M$$

$$2M = 4$$

$$M = 2$$

$$\text{or } M - \frac{1}{3} = -1 - \frac{M}{3}$$

$$3M - 1 = -3 - M$$

$$4M = -2$$

$$M = -\frac{1}{2} \quad \checkmark$$

$$e) \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{x/y + y/x + x/z}{x/y} \quad \checkmark$$

$$= \frac{-3}{-5} \quad \checkmark$$

$$= \frac{3}{5} \quad \checkmark$$

Done well

$$f) \int \sec 2u \tan 2u \, du = \frac{1}{2} \sec 2u + C$$

Calc - 2

Done well.

Q2 a) i) $\angle ACE = 36^\circ$ (angle between tangent and chord equals angle in alternate segment) $\checkmark$

Recs 3

Conn 3

Calc 2

ii) $\angle CAE = 90^\circ$ (angle in a semi-circle) $\checkmark$

$$\angle FAC = 180 - 90 - 36 \text{ (angle sum of a str. line)} \\ = 54^\circ$$

$\angle ADC = 54^\circ$ (angle between tangent and chord equals angle in alternate segment) $\checkmark$

Conn - 3

$$b) \int \frac{du}{\sqrt{25 - 4u^2}} = \frac{1}{2} \int \frac{du}{\sqrt{\frac{25}{4} - u^2}} \\ = \frac{1}{2} \sin^{-1} \left(\frac{2u}{5} \right) + C \quad \checkmark$$

Calc 2

$$c) i) \sin x - \sqrt{3} \cos x = R \sin x \cos \alpha - R \cos x \sin \alpha$$

$$\therefore R \sin \alpha = \sqrt{3} \dots \textcircled{1}$$

$$R \cos \alpha = 1 \dots \textcircled{2}$$

$$\textcircled{1} \div \textcircled{2} \quad \tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3} \quad \checkmark$$

$$\textcircled{1}^2 + \textcircled{2}^2 = R^2 \sin^2 \alpha + R^2 \cos^2 \alpha = (\sqrt{3})^2 + 1^2$$

$$R^2 = 4$$

$$R = 2 \quad \checkmark \text{ as } R > 0$$

$$\therefore \sin x - \sqrt{3} \cos x = 2 \sin \left(x - \frac{\pi}{3} \right)$$

$$\text{ii) } \sin x - \sqrt{3} \cos x = \sqrt{2}$$

$$2 \sin \left(x - \frac{\pi}{3} \right) = \sqrt{2}$$

$$\sin \left(x - \frac{\pi}{3} \right) = \frac{\sqrt{2}}{2}$$

$$x - \frac{\pi}{3} = n\pi + (-1)^n \frac{\pi}{4}$$

$$x = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{3} \quad \checkmark$$

$\checkmark$ for $\frac{\pi}{4}$

$$\text{d) prob of color blindness} = 0.05$$

$$\text{prob of not color blindness} = 0.95$$

$$\text{i) } P(X=0) = {}^{20}C_0 (0.05)^0 (0.95)^{20} \\ = 0.358 \quad \checkmark$$

$$\text{ii) } P(X \geq 2) = 1 - [P(X=0) + P(X=1)] \quad \checkmark \\ = 1 - \left[(0.95)^{20} + {}^{20}C_1 (0.05)(0.95)^{19} \right] \\ = 0.264 \quad \checkmark \text{ Rec-3}$$

$$\text{Q3 a) } \int_0^{3/2} \sqrt{9-x^2} dx \quad x = 3 \sin \theta \quad x = \frac{3}{2} \quad \theta = \frac{\pi}{6} \\ dx = 3 \cos \theta d\theta \quad x = 0 \quad \theta = 0$$

Rec-3

Conn-3

Calc-5

$$= \int_0^{\pi/6} \sqrt{9-9\sin^2 \theta} \cdot 3 \cos \theta d\theta \quad \checkmark$$

$$= \int_0^{\pi/6} 3 \cos \theta \cdot 3 \cos \theta d\theta$$

$$= 9 \int_0^{\pi/6} \cos^2 \theta d\theta$$

$$= \frac{9}{2} \int_0^{\pi/6} 1 + \cos 2\theta d\theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\cos^2 \theta = \frac{1}{2} (1 + \cos 2\theta)$$

This was done poorly by many students.

This is a standard integral type so you need to revise the work.

$$= \frac{9}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/6} \checkmark$$

$$= \frac{9}{2} \left[\left(\frac{\pi}{6} + \frac{1}{2} \sin \frac{\pi}{3} \right) - (0 + 0) \right]$$

$$= \frac{9}{2} \left(\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right)$$

$$= \frac{3\pi}{4} + \frac{9\sqrt{3}}{8} \checkmark$$

Calc-3

b) i) No of triangles = ${}^{12}C_3$
 $= 220 \checkmark$

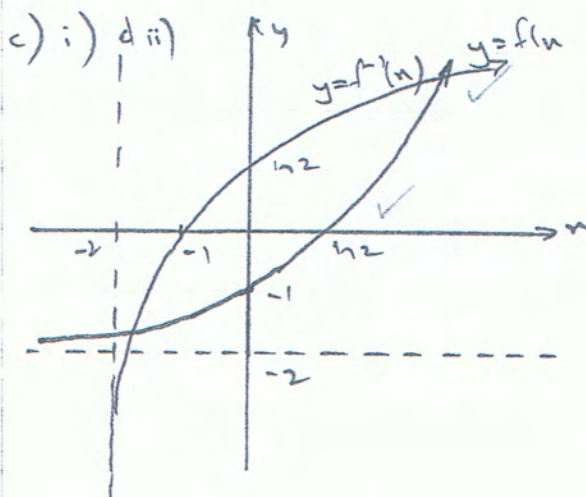
ii) No of triangles = ${}^7C_1 \times {}^4C_1 \times {}^1C_1$
 $= 28$

$$\therefore \text{Prob} = \frac{28}{220} = \frac{7}{55} \checkmark$$

iii) No of triangles = ${}^7C_3 + {}^4C_3$
 $= 35 + 4$
 $= 39$

$$\text{Prob} = \frac{39}{220} \checkmark$$

Reas-3



Conn-1

Conn-1

Done well

Some people didn't read the question regarding probability.

Done fairly well, although asymptotes were left out on many. Labeling x and y intercepts would have been nice

ii) $f(x): y = e^{x-2}$

interchange x and y

$$x = e^{y-2}$$

$$x+2 = e^y$$

$$y = \ln(x+2) \checkmark$$

Done well

iv) As the graph of $y = f^{-1}(x)$ is a reflection of the graph of $y = f(x)$ in the line $y = x$, points of intersection lie on $y = x$

$$\therefore \text{they satisfy } e^x - 2 = x \quad \checkmark$$

$$\therefore e^x - x - 2 = 0. \quad \text{Comm-1}$$

Done reasonably well
The key point is that $f(x)$ and $f^{-1}(x)$ intersect on $y = x$.

v) $f(x) = e^x - x - 2$

$$f'(x) = e^x - 1 \quad \checkmark$$

$$\therefore x_2 = 1.5 - \frac{f(1.5)}{f'(1.5)}$$

$$= 1.5 - \frac{e^{1.5} - 1.5 - 2}{e^{1.5} - 1}$$

$$= 1.22 \text{ (to 2 dec. pl.)} \quad \checkmark \quad \text{Calc 2}$$

Done very well

Q4 a) i) $x^2 = 8y$

Reas - 5

$$y = \frac{x^2}{8}$$

Comm - 2

$$\frac{dy}{dx} = \frac{x}{4}$$

Calc - 3

$$\therefore \text{at } P \quad M_{\text{tan}} = \frac{4p}{4} \quad \checkmark$$

$$= p.$$

$$\therefore M_{\text{norm}} = -\frac{1}{p}$$

$$\therefore y - 2p^2 = -\frac{1}{p}(x - 4p) \quad \checkmark$$

$$py - 2p^3 = -x + 4p$$

$$x + py = 2p^3 + 4p.$$

Comm - 2

ii) A: $x=0 \quad 0 + py = 2p^3 + 4p$

$$y = 2p^2 + 4$$

$$\therefore A(0, 2p^2 + 4) \quad \checkmark \quad P(4p, p^2)$$

$$\therefore \text{midpt } \left(\frac{0 + 4p}{2}, \frac{2p^2 + 4 + p^2}{2} \right) \quad \checkmark$$

$$= (2p, p^2 + 2). \quad \text{Reas - 2}$$

$$iii) x = 2p \quad y = 2p^2 + 2$$

$$p = \frac{x}{2} \text{ sub into } y$$

$$y = 2\left(\frac{x}{2}\right)^2 + 2 \checkmark$$

$$y = 2\frac{x^2}{4} + 2$$

$$4y = 2x^2 + 8$$

$$2y - 4 = x^2$$

$$x^2 = 2(y - 2)$$

$$\therefore 4a = 2 \quad a = \frac{1}{2} \quad \text{vertex } (0, 2) \checkmark$$

$$\text{focus } (0, 2\frac{1}{2}) \checkmark$$

Revs - 3

$$b) i) a = 3 \checkmark$$

$$b = \frac{1}{2} \checkmark$$

$$ii) y = 3 \cos^{-1} \frac{x}{2}$$

$$\frac{y}{3} = \cos^{-1} \frac{x}{2}$$

$$\frac{y}{3} = \cos^{-1} \frac{x}{2}$$

$$x = 2 \cos \frac{y}{3}$$

$$\therefore A = \int_0^{\pi/2} 2 \cos \frac{y}{3} dy \checkmark$$

$$= 2 \left[3 \sin \frac{y}{3} \right]_0^{\pi/2} \checkmark$$

$$= 6 \left(\sin \frac{\pi}{6} - \sin 0 \right)$$

$$= 6 \times \frac{1}{2}$$

$$= 3 \text{ units}^2 \checkmark$$

Calc - 3

$$QS a) i) V = Ah$$

$$= \frac{1}{2} \pi h \times 400$$

$$= 200\pi h$$

now by similar triangles

$$\frac{x}{h} = \frac{200}{25}$$

$$x = 8h \checkmark$$

$$\therefore V = 200 \times 8h \times h \checkmark$$

$$= 1600h^2$$

Revs - 2

Revs - 7

Comm - 2

Calc - 3

Similar triangle ques.
are quite popular so
for those who didn't
get this solution you
need to do further
practice.

$$\text{ii) } \frac{dV}{dt} = -16 \text{ L/s} \quad 1 \text{ L} = 1000 \text{ cm}^3$$

$$= -16000 \text{ cm}^3/\text{s}$$

$$\text{find } \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$\frac{dV}{dh} = 3200h \quad \checkmark$$

$$\therefore \frac{dh}{dt} = \frac{1}{3200h} \times -16000 \quad \checkmark$$

$$= -\frac{5}{h}$$

$$\therefore \text{ when } h = 10$$

$$\frac{dh}{dt} = -\frac{5}{10}$$

$$= -\frac{1}{2} \text{ cm/s} \quad \checkmark$$

Calc - 3

$$\text{b) } \text{LHS} = \frac{dT}{dt}$$

$$\text{RHS} = -k(T-8)$$

$$= -k(8 + 77e^{-kt} - 8)$$

$$= -k \times 77e^{-kt}$$

$$= -k \times 77e^{-kt} \quad \checkmark$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore T = 8 + 77e^{-kt} \quad \text{satisfies d.d. eqn.}$$

$$\text{when } t = 0 \quad T = 8 + 77e^{-k \times 0}$$

$$= 8 + 77 \quad \checkmark$$

$$= 85^\circ \text{C}$$

Comm - 2

$$\text{ii) } t = 60 \quad T = 40$$

$$\therefore 40 = 8 + 77e^{-k \times 60}$$

$$32 = 77e^{-k \times 60}$$

$$\frac{32}{77} = e^{-k \times 60}$$

$$-60k = \ln \frac{32}{77}$$

$$k = -\frac{\ln(32/77)}{60} \quad \checkmark$$

$$\text{find } t \text{ when } T = 10$$

Most students arrived

$$\text{at } \frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

but made a mistake with units

$$\text{i.e. } 16 \text{ L} = 16000 \text{ cm}^3$$

You can't mix units

in these questions

You need to be careful

with these questions

You need to have more

$$\text{than } \frac{dT}{dt} = -k77e^{-kt}$$

$$= -k(T-8)$$

initial conditions were

also missed by many.

Done very well.

$$\begin{aligned}
 10 &= 8 + 77e^{-kt} \\
 2 &= 77e^{-kt} \\
 \frac{2}{77} &= e^{-kt} \\
 -kt &= \ln \frac{2}{77} \\
 t &= \frac{\ln \left(\frac{2}{77} \right)}{-k} \quad \checkmark
 \end{aligned}$$

$$= 249.45 \dots \text{ mins}$$

$$= 4 \text{ hrs } 9 \text{ min}$$

$$1:09 \text{ pm.} \quad \checkmark$$

Revs 3

$$\begin{aligned}
 \text{c) coeff of } x^6 &: {}^{10}C_6 a^6 \\
 \text{coeff of } x^7 &: {}^{10}C_7 a^7 \\
 \therefore 210a^6 &= 2 \times 120a^7 \quad \checkmark
 \end{aligned}$$

$$a = \frac{210}{240}$$

$$= \frac{7}{8} \quad \checkmark$$

Revs - 2

Most common mistake
was $2 \times 710a^6 = 120a^7$

$$\begin{aligned}
 \text{Q6 a) i) } \frac{T_{n+1}}{T_n} &= \frac{{}^{20}C_n a^{20-n} b^n}{{}^{20}C_{n-1} a^{20-(n-1)} b^{n-1}} \\
 &= \frac{20!}{(20-n)!n!} \cdot \frac{a^{20-n} b^n}{a^{20-n+1} b^{n-1}} \\
 &= \frac{20!}{(20-(n-1))!(n-1)!} \cdot \frac{a^{20-n} b^n}{a^{20-n+1} b^{n-1}} \\
 &= \frac{b}{n} \cdot \frac{a}{20-(n-1)} \quad \checkmark
 \end{aligned}$$

Comm-2

$$= \frac{21-n}{n} \cdot \frac{b}{a}$$

$$\text{ii) } \alpha) P(7) = \frac{1}{6} \quad \checkmark$$

$$\beta) P(7) = {}^{20}C_n \left(\frac{5}{6} \right)^{20-n} \left(\frac{1}{6} \right)^n$$

$$\text{find } n \text{ such that } \frac{T_{n+1}}{T_n} > 1$$

$$\therefore \frac{21-n}{n} \cdot \frac{1}{\frac{5}{6}} > 1 \text{ from i).} \quad \checkmark$$

$$\frac{21-n}{5} \times \frac{1}{5} > 1$$

$$\frac{21-n}{5} > 1$$

$$21-n > 5n \quad \text{as } n > 0$$

$$6n < 21$$

$$n < \frac{21}{6}$$

$$\therefore n = 3 \quad \checkmark$$

$\therefore$ the most likely number of 7's thrown is 3

$$P(7) = {}^{20}C_3 \left(\frac{5}{6}\right)^{17} \left(\frac{1}{6}\right)^3 \quad \checkmark \quad \text{Recs} = 3$$

$$= 0.238 \quad (\text{to 3 d.p.})$$

b) i) show true for $n=1$

$$(1+n)^1 - 1 = 1+n-1$$

$$= n$$

which is divisible by n . $\checkmark$

assume true for $n=k$

$$\therefore (1+n)^k - 1 = Mn \quad \text{where } M \text{ is an integer}$$

show true for $n=k+1$

$$\text{i.e. } (1+n)^{k+1} - 1 = Qn \quad \text{where } Q \text{ is an integer}$$

$$\text{L.H.S.} = (1+n)^{k+1} - 1 = (1+n)(1+n)^k - 1 \quad \checkmark$$

$$= (1+n)(Mn+1) - 1 \quad \text{from above}$$

$$= Mn + 1 + Mn^2 + n - 1$$

$$= n(M + Mn + 1) \quad \checkmark$$

$$= Qn \quad \text{as } M \text{ and } n \text{ are}$$

positive integers

$\therefore$ result is true for $n=k+1$ Recs - 3

$\therefore$ we have shown that if it is true for $n=k$

then the result is true for $n=k+1$. The

result is true for $n=1$ $\therefore$ by the principle

of Mathematical Induction the result is true

for all positive n .

$$\begin{aligned} \text{ii) } 12^n - 4^n - 3^n + 1 &= 3^n \times 4^n - 4^n - 3^n + 1 \\ &= 4^n (3^n - 1) - 1(3^n - 1) \\ &= (3^n - 1)(4^n - 1) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{iii) } 12^n - 4^n - 3^n + 1 &= (3^n - 1)(4^n - 1) \\ &= ((2+1)^n - 1)((3+1)^n - 1) \quad \checkmark \end{aligned}$$

now $(2+1)^n - 1$ is divisible by 2 from i)

d $(3+1)^n - 1$ is divisible by 3 from i) $\checkmark$

$\therefore (3^n - 1)(4^n - 1)$ is divisible by $2 \times 3 = 6$.
Ans - 2

$$\begin{aligned} \text{Q7 a) i) } \left(x - \frac{1}{x}\right)^{2n} &= \sum_{r=0}^{2n} {}^{2n}C_r (x)^{2n-r} \left(-\frac{1}{x}\right)^r \\ &= \sum_{r=0}^{2n} {}^{2n}C_r (x)^{2n-r} (-x)^{-r} \quad \checkmark \end{aligned}$$

$\therefore$ for the term independent of x :

$$2n - r - r = 0 \quad \checkmark$$

$$2r = 2n$$

$$r = n.$$

$$\begin{aligned} \therefore \text{ term is } & {}^{2n}C_n (x)^{2n-n} (-x)^{-n} \\ &= {}^{2n}C_n (-1)^n \quad \text{Ans - 2} \end{aligned}$$

$$\begin{aligned} \text{ii) } \text{LHS} &= \left(1+x\right)^{2n} \left(1-\frac{1}{x}\right)^{2n} = \left[\left(1+x\right)\left(1-\frac{1}{x}\right)\right]^{2n} \\ &= \left(1-\frac{1}{x}+x-1\right)^{2n} \quad \checkmark \\ &= \left(x-\frac{1}{x}\right)^{2n} \quad \text{Conn-1} \\ &= \text{RHS.} \end{aligned}$$

$$\begin{aligned} \text{iii) } & \text{expand } \left(1+x\right)^{2n} \left(1-\frac{1}{x}\right)^{2n} \\ &= \left({}^{2n}C_0 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_{2n} x^{2n}\right) \\ & \quad \times \left({}^{2n}C_0 - {}^{2n}C_1 \left(\frac{1}{x}\right) + {}^{2n}C_2 \left(\frac{1}{x^2}\right) - \dots + {}^{2n}C_{2n} \frac{1}{x^{2n}}\right) \end{aligned}$$

$$= \dots + {}^{2n}C_0 \times {}^{2n}C_0 - {}^{2n}C_1 x {}^{2n}C_1 \left(\frac{1}{x}\right) + \dots + {}^{2n}C_{2n} x^{2n} \times {}^{2n}C_{2n} \frac{1}{x^{2n}} + \dots \quad \checkmark$$

$\therefore$ terms independent of x in this expansion

are:

$$\left({}^{2n}C_0\right)^2 - \left({}^{2n}C_1\right)^2 + \left({}^{2n}C_2\right)^2 - \dots + \left({}^{2n}C_{2n}\right)^2$$

Done fairly well

Easy mark thrown away by many.

Just because this question is complicated there are still easy parts.

Many ~~for the~~ students didn't know which terms to multiply together.

$$\text{now } \left(1+\frac{1}{n}\right)^{2n} \left(1-\frac{1}{n}\right)^{2n} = \left(n-\frac{1}{n}\right)^{2n}$$

and the term independent of x in $\left(n-\frac{1}{n}\right)^{2n}$ is $(-1)^n {}^{2n}C_n$ ✓

∴ equating terms

Recall - 2

$$\left({}^{2n}C_0\right)^2 - \left({}^{2n}C_1\right)^2 + \left({}^{2n}C_2\right)^2 - \dots + \left({}^{2n}C_{2n}\right)^2 = (-1)^n {}^{2n}C_n$$

b) i) In $\triangle AXB$ and $\triangle DXC$

$$\frac{AX}{DX} = \frac{XB}{XC} \quad (\because AX \cdot XC = BX \cdot XD) \quad \checkmark$$

$$\angle BXA = \angle CXD \quad (\text{vert. opp angles are eq.}) \quad \checkmark$$

∴ $\triangle AXB \parallel \triangle DXC$ (two pairs of sides in the same ratio and the included angle is equal) ✓

$$\therefore \angle ABX = \angle DCX \quad (\text{corr. angles in similar triangles})$$

∴ ABCD is a cyclic quadrilateral as

angles in the same segment are equal. ✓

Comm - 4

ii) a) B and C are x-intercepts ∴ $y=0$

$$\therefore x^2 + px - q = 0$$

$$x = \frac{-p \pm \sqrt{p^2 - 4 \cdot 1 \cdot -q}}{2 \times 1}$$

$$= \frac{-p \pm \sqrt{p^2 + 4q}}{2}$$

$$\therefore B\left(\frac{-p - \sqrt{p^2 + 4q}}{2}, 0\right) \text{ \& \& } C\left(\frac{-p + \sqrt{p^2 + 4q}}{2}, 0\right) \quad \checkmark$$

$$\therefore OB = \left| \frac{-p - \sqrt{p^2 + 4q}}{2} \right| \quad OC = \left| \frac{-p + \sqrt{p^2 + 4q}}{2} \right|$$

$$= \frac{p + \sqrt{p^2 + 4q}}{2} \quad = \frac{-p + \sqrt{p^2 + 4q}}{2}$$

$$\therefore OB \times OC = \frac{p + \sqrt{p^2 + 4q}}{2} \times \frac{-p + \sqrt{p^2 + 4q}}{2}$$

$$= \frac{-p^2 + p\sqrt{p^2 + 4q} - p\sqrt{p^2 + 4q} + (\sqrt{p^2 + 4q})^2}{4}$$

Most students who attempted this question

✓ knew the concepts involved but many struggled with presenting a formal and logical proof.

Done well for those students who knew what to do!

Be careful not OB and OC are lengths ∴ $OB = \left| \frac{-p + \sqrt{p^2 + 4q}}{2} \right|$

$$= \frac{-p^2 + p^2 + 4q}{4} \checkmark$$

$$= q \checkmark$$

$$OA = |q|$$

$$OD = |1|$$

$$= q \checkmark$$

$$= 1$$

$$\therefore OA \times OD = q \times 1$$

$$= q \checkmark$$

$$\therefore OB \times OC = OA \times OD$$

Revs-3

$\therefore ABDC$ is a cyclic quad.